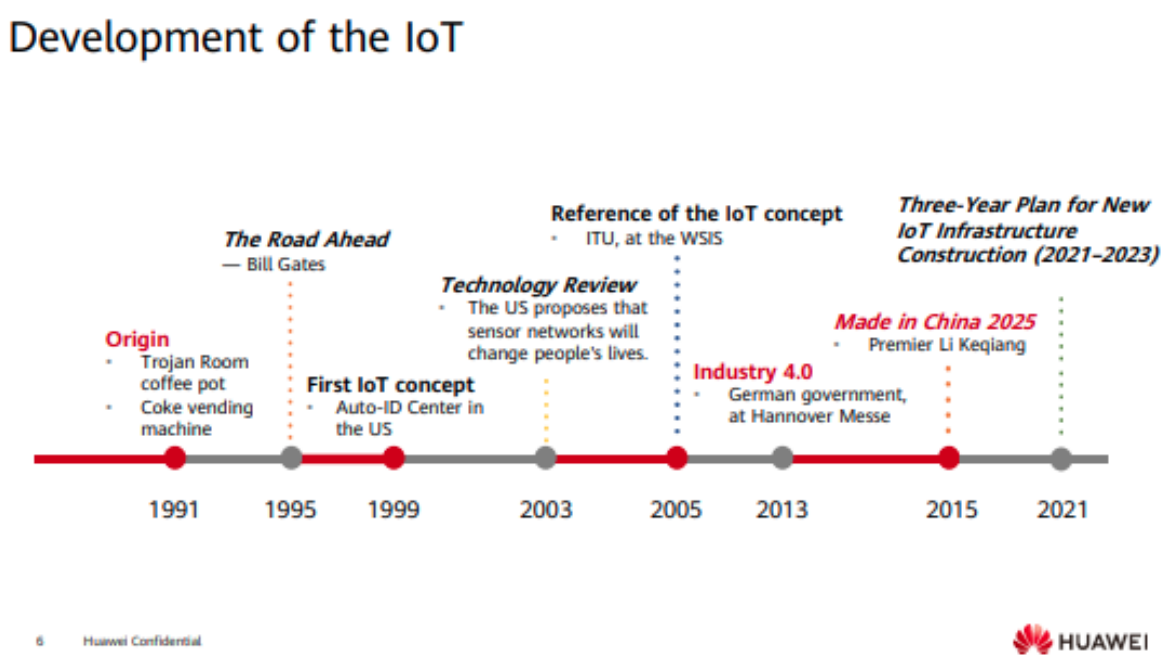
Chapter 1: A History and Overview of IoT

1.1 IoT Development History

Origin of the IoT

Trojan Room coffee pot in 1991: At the computer Laboratory in Cambridge University, scientists would go downstairs only to find that the coffee was not ready.

To solve this problem, they wrote a program, installed a portable camera next to the coffee pot and used image capture technology to check whether the coffee was ready. They monitored the coffee pot using a camera eliminating the need to go up and down the stairs.



• The concept of the IoT dates back to **Bill Gates's book, The Road Ahead 1995.** However, due to the limited development of wireless networks, hardware, and sensors, his idea did not attract much attention.

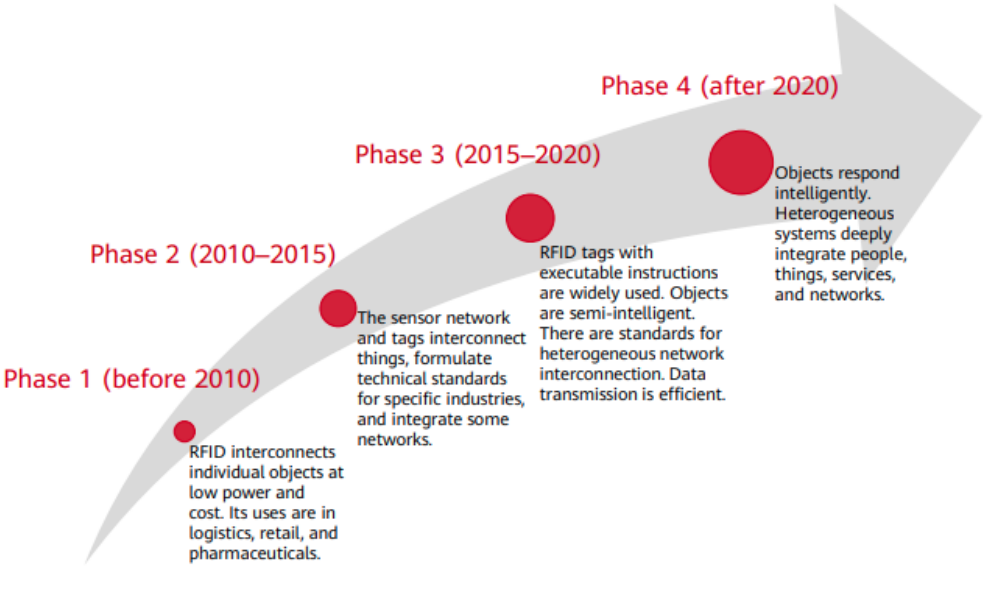
• November 17, 2005, the International Telecommunication Union (ITU) released the Internet Reports 2005: The Internet of Things at the World Summit on the Information Society (WSIS).

• August 7, 2009, China's former Premier Wen Jiabao delivered an important

speech during the inspection in Wuxi city, proposing the "Sensing China" strategy and saying that China should seize the opportunity and vigorously develop IoT technologies. November 3 in the same year, he delivered a speech to the science and technology circle of Beijing.

• **Smart Logistics, proposed in Industry 4.0, integrates logistics resources through the Internet, IoT, and logistics networks**, to maximize the efficiency of existing logistics resource suppliers.

EU and National: IoT Industry Development Plans



- South Korea's government proposes the concept of ubiquitous sensing network.
- The IT Strategy Department of the Japanese government has formulated a nextgeneration informatization strategy
- In December 2021, LoRaWAN was recognized by the International Telecommunication Union (ITU) as standard. The approval affirms the IoT achievements and prospects of LoRa and its ecosystem.
- The ITU-T Recommendation Y.4480: Low Power Protocol for Wide Area Wireless Networks is delivered by the Study Group 20 of International Telecommunication Union-Telecommunication Standardization Sector (ITU-T)

and was approved by ITU. LoRaWAN has officially become a global standard, boosting the IoT industry.

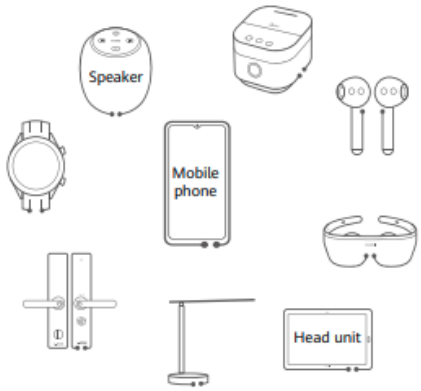
IoT Key Event: NB-IoT Officially Incorporated into 5G Standards

ITU-R WP5D meeting on July 9, 2020: NB-IoT and NR are now 5G standards. So far, NB-IoT has been officially defined as the cornerstone for building a fully connected, intelligent world.

• NB-IoT can be deployed in NR in-band mode to support access to the 5G core network. NB-IoT 3GPP specifications will evolve and become the core standard for 5G mMTC scenarios. 5G mMTC requires 1 million device connections per square kilometer. NB-IoT can provide 100,000 connections with its 200 kHz frequency. NB-IoT does meet ITU's requirements for low power consumption and wide 5G coverage.

• 5G can better satisfy e-commerce industries in terms of IoT development and help them to go digital. NB-IoT is designed for the communication between things. In recent years, NB-IoT has been widely used, improving and even changing the production and operations of some industries. With NB-IoT applications, smart parking, smart firefighting, smart water, smart street lamps, bicycle sharing, and smart home appliances have greatly changed consumer behavior. NB-IoT is a stepping stone for 5G to enter the e-commerce industry.

IoT Industry Status: Consumer IoT

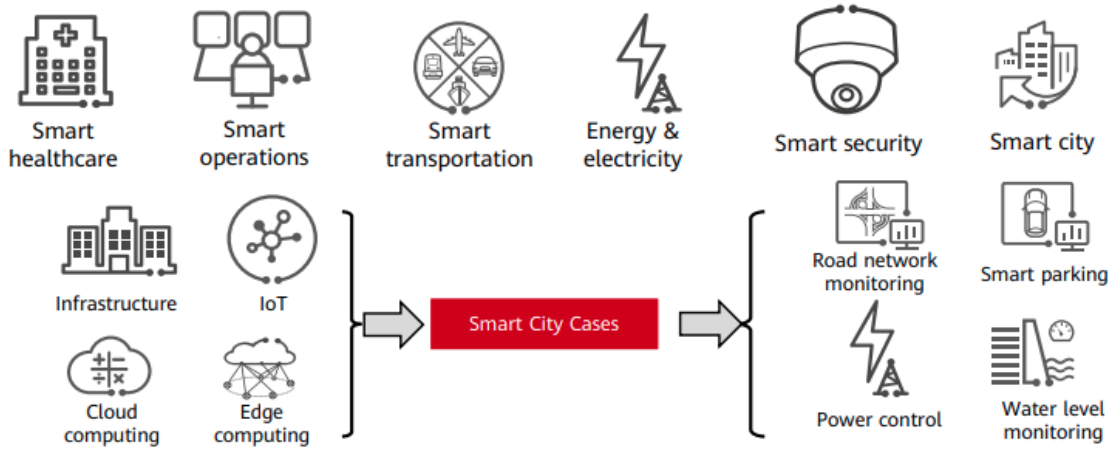


Consumer IoT: Products and services that we directly buy and use in all aspects of life. Examples: Smart bands, VR/AR glasses, body fat scales, smart locks, smart speakers, shared bikes/electric bikes, and autonomous driving. MarketsandMarkets reports that the global consumer IoT market was USD46.8 billion in 2018 and will reach USD104.4 billion by 2023. The compound annual growth rate (CAGR) from 2018 to 2023 is 17.39%.

There are consumer IoT and industry IoT. Industry IoT is the combination of policy-driven IoT and production IoT.

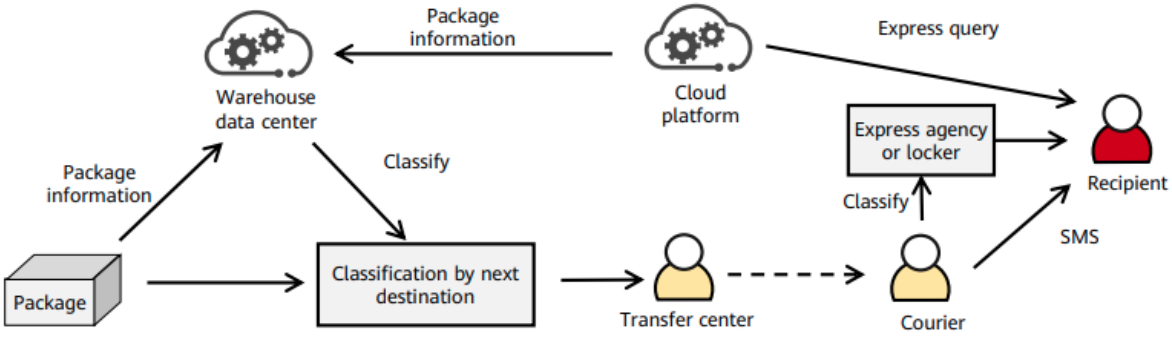
IoT Industry Status: Policy-Driven IoT

- Mainly used for city management (fire fighting, security, system integration, public utilities, lighting, and parking).

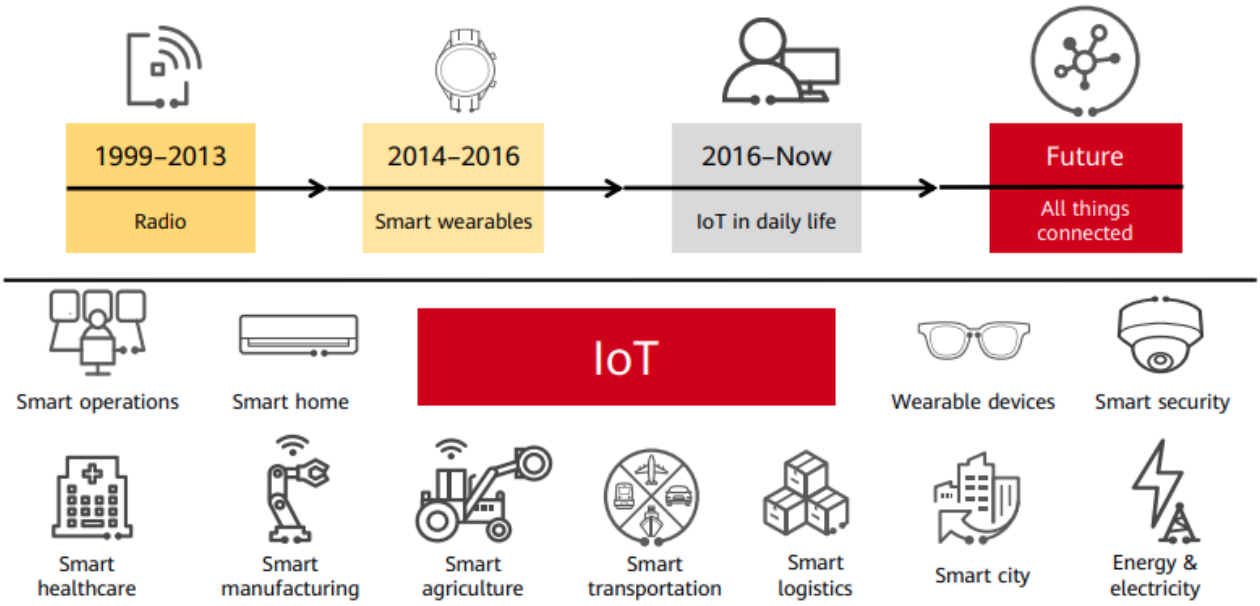


IoT Industry Status: Production IoT

- Covers smart industry, Internet of Vehicles (IoV), smart logistics, and smart agriculture. For example, an IoT application checks the status of a package periodically and uploads this data to the cloud.



Application and Development of IoT



1.2 IoT Overview and Architecture

IoT or Internet of Things Refers to the concept where physical objects or devices are connected to the internet and can communicate with each other. The main idea behind IoT is to give everyday objects the ability to collect and share data with other devices over an internet connection.

In its implementation, IoT devices are usually equipped with sensors, software, and network connectivity to collect data, communicate with other devices, and/or take appropriate actions based on the collected data. Examples of IoT devices include smart sensors, wearable devices, smart lights, smart security cameras, and more.

IoT has the potential to change the way we interact with our surroundings and enable innovative solutions in areas such as health, transportation, industry and smart homes. However, successful IoT implementation also brings challenges related to data security, privacy, and device interoperability.

Application

It is a place where data collected by IoT devices is used to provide specific solutions or run specific applications. It covers various types of applications, such as smart home, smart health, smart manufacturing and others.

Its function is to use data collected from the device to provide the desired solution, provide information to the end user, or control the device automatically.

Platform

Providing hardware and software infrastructure that supports IoT device operations and management. This includes data storage, cloud computing, device management, and other related services.

Its function is to provide the basis for storing, processing and managing data from IoT devices. Establishment of integration with applications and provision of services necessary to support IoT functionality.

Network

Includes the infrastructure that allows IoT devices to communicate with each other and with backend systems. This includes communication protocols, wireless connections, and other network elements.

Its function is to provide the necessary communication paths between IoT devices. transfer data reliably and securely between devices and backends.

Sensing

Sensing includes physical devices or sensors that capture data from the surrounding environment. These sensors can measure various parameters such as temperature, humidity, light, movement, and others.

Its function is to collect data from the surrounding environment and transmit it to other layers in the IoT architecture. This data is then used for analysis, decision making, or automated actions by devices or systems.

1.3 Huawei IoT Solution

• One IoT platform, two access methods, and one IoT operating system

• A secure, reliable platform that supports decoupling from devices and applications is required to develop IoT services.

• The Huawei Cloud IoT platform provides device access, security verification, service orchestration, data management, and multi-protocol communications, implements connectivity management and capability openness, and connects to carriers' pipes and platforms.

• NB-IoT: Narrowband Internet of Things

• NB-IoT is a new narrowband cellular communications LPWAN technology. LPWAN stands for low-power wide-area network. NB-IoT has strong connection, wide coverage, low cost, and low power consumption.

• CoAP: Constrained Application Protocol

• MQTT: Message Queuing Telemetry Transport

• SOAP: Simple Object Access Protocol

• HTTP: Hypertext Transfer Protocol

• Network function virtualization (NFV): Uses universal hardware (such as x86) and virtualization to process software with many functions, reducing network device costs. Using software and hardware decoupling and function abstraction, it enables network device functions to be independent of dedicated hardware. Resources can be flexibly shared, new services can be quickly developed and deployed, and automated deployment, auto scaling, fault isolation, and selfhealing can be performed as required.

• Software-defined networking (SDN): An innovative network architecture proposed by the Clean Slate Program research team from Stanford University. It is an approach to implement network virtualization. Its core technology, OpenFlow, separates the control plane from the data plane of a network device, thereby network traffic can be flexibly controlled. This makes the network pipe more intelligent and provides a good platform for innovation of the core network and applications.

• Network slicing virtualizes a physical network into multiple end-to-end (E2E) networks, which provide different functions to meet different service requirements.

• Network slices have three key features: customization, end to end, and isolation.

    ▫ A network slice is an end-to-end network, covering the RAN, transport network, and core network. It requires a cross-domain slice management system.

    ▫ Network slices require isolation of resources, security, and OAM. Different domains can use different technologies to achieve such isolation. For

example, the core network uses virtualization technology.

▫ Network slices can be customized to provide specific network functions and features.

• The industrial IoT gateway supports multiple network standards and provides interfaces that comply with RS-485, PLC, and ZigBee. The gateway can directly communicate with devices on the sensor network, converge heterogeneous sensor networks that adopt different technologies, and remotely transmit sensor data using communications networks.

• The industrial IoT gateway functions just like a common Internet router and connects to the 3G mobile network and Internet in wireless or wired mode. The industrial IoT gateway can also interconnect with carriers' platforms to provide better services.

• The industrial IoT gateway supports Quality of Service (QoS) to ensure that each area or application corresponds to an independent queue, so applications can be scheduled in real time. It also provides dedicated bandwidth and preferential forwarding for key services to secure the service quality.

• The industrial IoT gateway provides security mechanisms to ensure authenticity and accuracy of IoT big data. In addition, the gateway devices must be industrialized, support a wide temperature range, and be dustproof, waterproof, and strong electromagnetic-proof to adapt to different scenarios.

• Carriers are in urgent need of a product that can function as an entry to access home networks to realize smart home. That is why smart ONT comes into being.
Smart ONT has one core, two open advanced architectures, and a variety of innovative business models. Smart ONT provides smart full Wi-Fi and IoT connection extended by USB dongles to achieve ubiquitous connections. It extends flexible pipes from carriers' equipment rooms to users' homes, integrates applications, and builds a platform for smart home. Smart ONTs delivers security, home control, energy management, distance education, and telemedicine to help carriers expand the home market. The advantages of smart ONTs in smart connectivity, intelligent services, and intelligent O&M help carriers increase revenue and reduce expenditure.

• Huawei LiteOS is a unified IoT OS with middleware. It is lightweight (basic kernel size less than

10 KB), low power consumption, interconnected, and secure.

- Currently, Huawei LiteOS is used on smart hardware in smart home, wearable devices, IoV, smart metering, and industrial Internet. It can also connect to LiteOS hardware to improve user experience.
    - The LiteOS features low power consumption, small kernels, and quick response.

In addition, the LiteOS has established an open source community to support chipsets such as the HiSilicon PLC chipset HCT3911, media chipset 3798M and 3798C, IP camera chipset Hi3516A, and LTE-M chipset.

1.4 IoT Engineering and Engineers

- Following the "Platform+Ecosystem" development strategy, Huawei Certification is a new collaborative architecture of ICT infrastructure based on "Cloud-Pipe-Device". Huawei has set up a complete certification system consisting of ICT infrastructure certification and platform and service certification. It is the only certification system that covers all ICT technical fields in the industry.
- HCIA-IoT Training and Certification aims to train and certify engineers who are able to develop E2E IoT services based on the Huawei IoT solution architecture.
- Engineers certified by HCIA-IoT master basic IoT knowledge and the Huawei IoT solution architecture, and are able to develop E2E services and perform O&M based on this architecture.

- The IoT is widely used in smart home, smart wearables, smart city, and smart agriculture. Currently, commercial IoT is craving for professionals and the payment is handsome. Still, you need to earn your way through hard work.

- The main tasks of IoT engineers of the Ministry of Human Resources and Social Security can be matched with the course content of IoT certification.

QUIZ

1. In the construction of a smart city, the data of many devices is stored in a unified manner at which layer?
   A. Sensing Layer
   B. Network Layer
   C. Platform Layer
   D. Application Layer

2. What type of IoT drive is the smart logistics case?
   A. Consumer IoT
   B. Policy-Driven IoT
   C. Production IoT
   D. None

3. Which of the following is “2” in Huawei’s “1+2+1” IoT solution architecture?
   A. IoT platform
   B. Home Gateway
   C. NB-IoT
   D. LiteOS

4. What are the layers of the IoT architecture?
   A. Sensing Layer
   B. Network Layer
   C. Platform Layer
   D. Application Layer

5. IoT is the internet where things are interconnected. IoT devices can be connected only through wireless network
   A. Right
   B. Wrong